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Deploying and Using vJunos in a Bare Metal EVE-NG server

An HPE Juniper Networking Techpost

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1 Deploying and Using vJunos in a Bare Metal EVE-NG server

Shalini Mukherjee - 05/11/2023

vJunos-switch and vJunosEvolved deployed on EVE-NG and integrated with Juniper Apstra to build a complete Data Center fabric. Article co-written with Aninda Chatterjee and Shalini Mukherjee, TME in Juniper Networks Cloud-Ready Data Center team.

1.1 Introduction

In this article, we will look at how vJunos-switch and vJunosEvolved is deployed on a bare metal install of EVE-NG. This deployment will then be used to integrate with Juniper Apstra and build a complete Data Center fabric.

We'll start with instructions on how to deploy EVE-NG as a bare metal install on an Ubuntu server and run through the EVE-NG installation itself as an example of how this can be done. Once EVE-NG is setup, we'll show how vJunos-switch and vJunosEvolved can be added to EVE-NG as a deployable node and the EVE-NG template that is needed for this to work.

Finally, we'll wrap this up by onboarding vJunos-switch and vJunosEvolved devices in Juniper Apstra and building a simple Data Center fabric to show that these devices work seamlessly with Apstra as well.

> Note that another post on vJunos installation on KVM has been published too.

1.1.1 What is EVE-NG and vJunos-switch/vJunosEvolved?

EVE-NG is a popular network emulation software that offers multivendor support. It allows a user to build network topologies on a blank canvas, providing means to test and simulate various network technologies and features across different network operating systems and products.

EVE-NG has various deployment options:

- as a VM on top of a hypervisor like ESXi
- over a bare metal server
- cloud deployment options (like GCP)

vJunos-switch and vJunosEvolved are two new virtual software offerings from Juniper Networks. vJunos-switch emulates the software functionality of Junos OS while vJunosEvolved emulates the software functionality of Junos EVO.

Since vJunos-switch is only supported on a bare metal install of EVE-NG, that is the deployment option we'll be going through in this blog post.

1.2 Installing EVE-NG on a Bare Metal Ubuntu Server

EVE-NG comes in two flavors: a community (free) edition, and a professional (paid) edition. We'll take a look at how to do a bare metal EVE-NG community edition install in this post, since that is more complicated. The .iso file for the community edition can be downloaded from the EVE-NG website - https://www.eve-ng.net/index.php/download/

Once this file is downloaded, mount the .iso file on your local machine. For example, on a MAC, you can simply double click the file to mount it. This should create a new drive, whose contents can be accessed now:

```
anindac@ubuntu EVE-NG Community % ls -l
total 216
dr-xr-xr-x 3 aninchat staff 2048 Feb 23 2022 EFI
dr-xr-xr-x 3 aninchat staff 2048 Feb 23 2022 boot
dr-xr-xr-x 3 aninchat staff 2048 Feb 23 2022 casper
dr-xr-xr-x 3 aninchat staff 2048 Feb 23 2022 dists
dr-xr-xr-x 2 aninchat staff 2048 Feb 23 2022 install
dr-xr-xr-x 2 aninchat staff 36864 Feb 23 2022 isolinux
-r--r--r-- 1 aninchat staff 27257 May 21 2022 md5sum.txt
-r--r--r-- 1 aninchat staff 27389 Feb 23 2022 md5sum.txt.orig
dr-xr-xr-x 3 aninchat staff 2048 Feb 23 2022 pool
dr-xr-xr-x 2 aninchat staff 2048 Feb 23 2022 preseed
dr-xr-xr-x 2 aninchat staff 2048 May 20 2022 server
```

Here, should see a folder titled ‘server’. This has a shell script that installs EVE-NG as a bare metal install. The script is as follows:

```
anindac@ubuntu EVE-NG Community % cd server
anindac@ubuntu server % ls -l
total 8
-r-xr-xr-x 1 aninchat staff 802 May 20 2022 eve-setup.sh
-r--r--r-- 1 aninchat staff 0 May 14 2022 meta-data
-r--r--r-- 1 aninchat staff 956 May 20 2022 user-data

anindac@ubuntu server % cat eve-setup.sh
#!/bin/sh
#Modify /etc/ssh/sshd_config with: PermitRootLogin yes
sed -i -e "s/.*PermitRootLogin .*/PermitRootLogin yes/" /etc/ssh/sshd_config
sed -i -e 's/.*DefaultTimeoutStopSec=.*/DefaultTimeoutStopSec=5s/' /etc/systemd/system.conf
systemctl restart ssh
apt-get update
apt-get -y install software-properties-common
wget -O - http://www.eve-ng.net/focal/eczema@ecze.com.gpg.key | sudo apt-key add -
sudo add-apt-repository "deb [arch=amd64] http://www.eve-ng.net/focal focal main"
apt-get update
DEBIAN_FRONTEND=noninteractive apt-get -y install eve-ng
/etc/init.d/mysql restart
DEBIAN_FRONTEND=noninteractive apt-get -y install eve-ng
echo root:eve | chpasswd
ROOTLV=$(mount | grep ' / ' | awk '{print $1}')
echo $ROOTLV
lvextend -l +100%FREE $ROOTLV
echo Resizing ROOT FS
resize2fs $ROOTLV
reboot
```

On your Ubuntu 20.04 server, simply run through these steps one by one. Once you reboot (the final step of the script), and the server comes back online, you should be able to SSH into the device with the default credentials of root/eve.

Once you’ve logged in, the actual EVE-NG installation is kicked off – this includes providing a hostname, IP address and gateway, primary and secondary DNS servers, NTP servers and so on. When this finishes, EVE-NG should be up and running, and accessible via both the CLI and UI (default credentials for UI login is admin/eve).

To deploy the professional version of EVE-NG, similar steps can be followed. Alternatively, the .iso file can be mounted directly as a virtual drive and you can boot from it to start both the Ubuntu and the EVE-NG installation together.

1.2.1 Deploying vJunos-switch and vJunosEvolved in EVE-NG

EVE-NG uses templates to boot various operating systems it supports – these templates are pre-built and come packaged within the installation of EVE-NG. They contain instructions on how to boot the devices, including how many interfaces to assign to the device, the kind of CPU/memory the device needs and so on.

The templates are stored in the path '*/opt/unetlab/html/templates/intel/*' and are written in YAML format, as seen below:

```
root@eve-ng:/opt/unetlab/html/templates/intel# ls -l
total 588
-rw-r--r-- 1 root root 10 Jan 31 07:15 '*.yml'
-rw-r--r-- 1 root root 1944 May 14 2022 128T.yml
-rw-r--r-- 1 root root 1853 May 14 2022 a10.yml
-rw-r--r-- 1 root root 1897 May 14 2022 acs.yml
-rw-r--r-- 1 root root 1913 May 14 2022 alienvault.yml
-rw-r--r-- 1 root root 1911 May 14 2022 alteon.yml
-rw-r--r-- 1 root root 1848 May 14 2022 ampcloud.yml
-rw-r--r-- 1 root root 1885 May 14 2022 android.yml
-rw-r--r-- 1 root root 1921 Jan 31 07:15 aos.yml
-rw-r--r-- 1 root root 1890 May 14 2022 apicem.yml
-rw-r--r-- 1 root root 1952 May 14 2022 apstra-aos.yml
-rw-r--r-- 1 root root 1951 May 14 2022 apstra-ztp.yml
-rw-r--r-- 1 root root 1950 May 14 2022 aruba.yml
-rw-r--r-- 1 root root 1919 May 14 2022 arubaamp.yml
-rw-r--r-- 1 root root 2002 May 14 2022 arubacx.yml
-rw-r--r-- 1 root root 1930 May 14 2022 arubanet.yml
-rw-r--r-- 1 root root 1971 May 14 2022 asa.yml
-rw-r--r-- 1 root root 1950 May 14 2022 asav.yml
-rw-r--r-- 1 root root 1899 May 14 2022 barracuda.yml
-rw-r--r-- 1 root root 1930 May 14 2022 bigip.yml
-rw-r--r-- 1 root root 1915 May 14 2022 brocadevadx.yml
-rw-r--r-- 1 root root 1829 May 14 2022 c1710.yml
-rw-r--r-- 1 root root 1931 May 14 2022 c3725.yml
-rw-r--r-- 1 root root 1993 May 14 2022 c7200.yml
-rw-r--r-- 1 root root 2017 May 14 2022 c8000v.yml
-rw-r--r-- 1 root root 1944 May 14 2022 c9800cl.yml
-rw-r--r-- 1 root root 1839 May 14 2022 cda.yml
-rw-r--r-- 1 root root 1981 May 14 2022 cexpresw.yml
-rw-r--r-- 1 root root 1973 May 14 2022 cips.yml
-rw-r--r-- 1 root root 1974 Jul 8 2022 clavisterc.yml
-rw-r--r-- 1 root root 1925 May 14 2022 clearpass.yml
-rw-r--r-- 1 root root 1917 May 14 2022 cms.yml
-rw-r--r-- 1 root root 1862 May 14 2022 coeus.yml
-rw-r--r-- 1 root root 1935 May 14 2022 cpsg.yml
-rw-r--r-- 1 root root 1938 May 14 2022 csr1000v.yml
-rw-r--r-- 1 root root 1967 May 14 2022 csr1000vng.yml
-rw-r--r-- 1 root root 1997 May 14 2022 ctxsdw.yml
-rw-r--r-- 1 root root 1966 May 14 2022 cuc.yml
-rw-r--r-- 1 root root 1931 May 14 2022 cucm.yml
-rw-r--r-- 1 root root 1909 May 14 2022 cue.yml
-rw-r--r-- 1 root root 1870 May 14 2022 cumulus.yml
-rw-r--r-- 1 root root 1998 May 14 2022 cup.yml
-rw-r--r-- 1 root root 1955 May 14 2022 cvp.yml
-rw-r--r-- 1 root root 1878 May 14 2022 cyberoam.yml
-rw-r--r-- 1 root root 1957 May 14 2022 dcnm.yml
-rw-r--r-- 1 root root 1969 May 14 2022 dellos10.yml
-rw-r--r-- 1 root root 1769 May 14 2022 docker.yml
-rw-r--r-- 1 root root 1920 May 14 2022 esxi.yml
-rw-r--r-- 1 root root 1862 May 14 2022 extremexos.yml
-rw-r--r-- 1 root root 1898 May 14 2022 firepower.yml

*snip*
```

Since vJunos-switch and vJunosEvolved are entirely new offerings, there is no pre-built template for this. The template must be created manually. The following templates can be used for vJunos-switch and vJunosEvolved:

vJunosEvolved:

```
root@eve-ng:/opt/unetlab/html/templates/intel# cat vJunosEVO.yml
```

```
type: qemu
description: vJunos-switch
name: vJunos-switch
cpulimit: 4
icon: JunipervQFXre.png
cpu: 4
ram: 5120
eth_name:
- fxp0
- ge-0/0/0
- ge-0/0/1
- ge-0/0/2
- ge-0/0/3
- ge-0/0/4
- ge-0/0/5
- ge-0/0/6
- ge-0/0/7
- ge-0/0/8
- ge-0/0/9
ethernet: 11
console: telnet
qemu_arch: x86_64
qemu_version: 4.1.0
qemu_nic: virtio-net-pci
qemu_options: -machine type=pc,accel=kvm -serial mon:stdio -nographic -smbios type=1,product=VM-VEX -cpu IvyBridge,ibpb=on,md-clear=on,spec-ctrl=on,ssbd=on,vmx=on
```

The next step is to create a new directory for this device and copy the image to this directory. We'll create a directory with the prefix 'vJunosEVO' and 'vJunos-switch' followed by a suffix which specifies the version of the software image. The prefix naming convention is important – it must match the name of the template that exists for the device. EVE-NG uses this name to determine which template must be used to boot the device (hence, the need for them to match).

The images (and their directories) are stored under '*/opt/unetlab/addons/qemu/*'.

```
root@eve-ng:/opt/unetlab/addons/qemu# pwd
/opt/unetlab/addons/qemu
root@eve-ng:/opt/unetlab/addons/qemu# mkdir vJunosEVO-23.2R1

root@eve-ng:/opt/unetlab/addons/qemu# cd vJunosEVO-23.2R1/
root@eve-ng:/opt/unetlab/addons/qemu/vJunosEVO-23.2R1# ls -l
total 1714308
-rw-r--r-- 1 root root 1755447296 Mar 14 00:14 virtioa.qcow2
```

and

```
root@eve-ng:/opt/unetlab/addons/qemu# pwd
/opt/unetlab/addons/qemu
root@eve-ng:/opt/unetlab/addons/qemu# mkdir vJunosswitch-23.2R1

root@eve-ng:/opt/unetlab/addons/qemu# cd vJunosswitch-23.2R1/
root@eve-ng:/opt/unetlab/addons/qemu/vJunosswitch-23.2R1# ls -l
total 3850436
-rw-r--r-- 1 root root 3942842368 Feb 25 11:15 vjunos-switch-23.2R1.qcow2
```

Once the image is copied into the folder, it must be renamed to 'virtioa.qcow2' as per EVE-NGs naming convention. Finally, EVE-NG requires you to fix some permissions to use the image - they have a pre-built script for this which can be invoked using '*/opt/unetlab/wrappers/unl_wrapper -a fixpermissions*' as seen below:

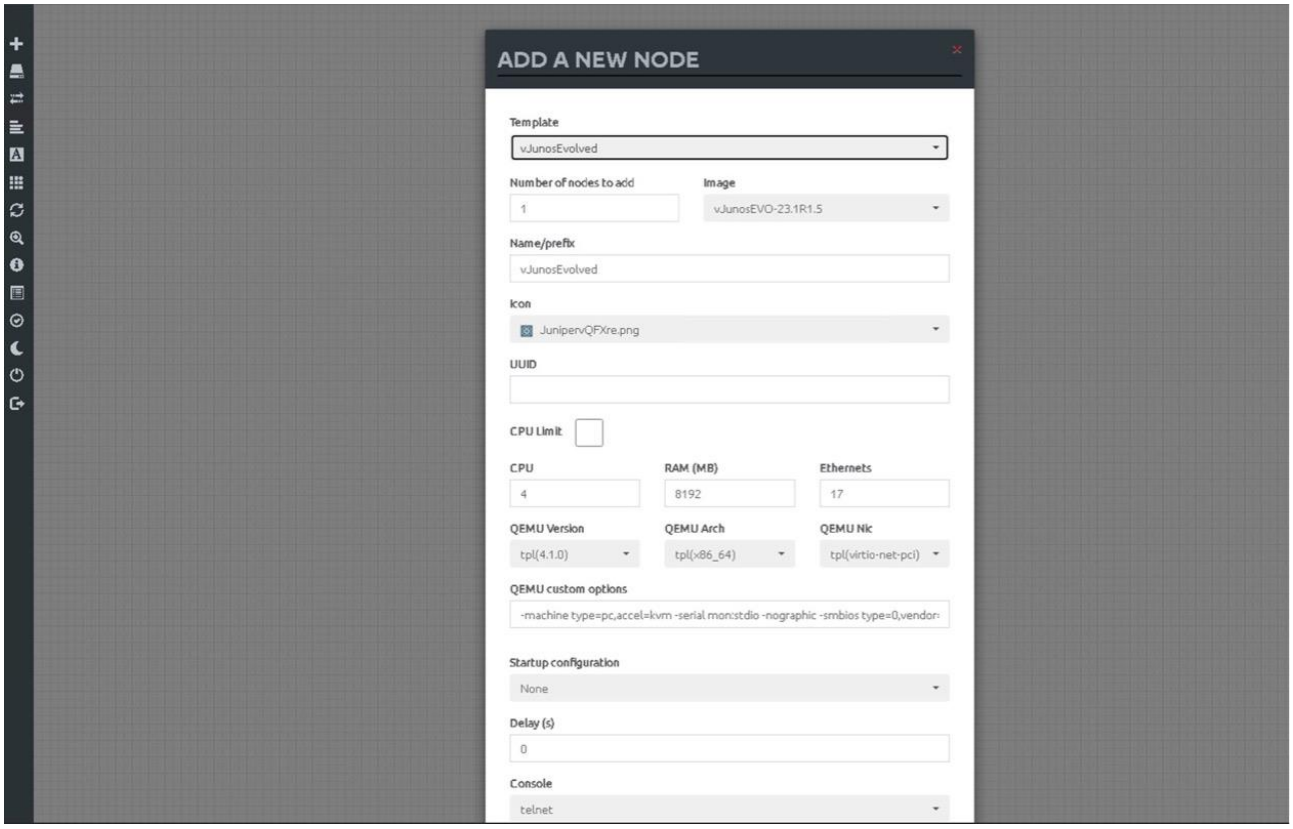
```
root@eve-ng:/opt/unetlab/addons/qemu/vJunosEVO-23.2R1# mv vjunosEVO-23.2R1.qcow2 virtioa.qcow2
root@eve-ng:/opt/unetlab/addons/qemu/vJunosEVO-23.2R1# /opt/unetlab/wrappers/unl_wrapper -a fixpermissions
```

and

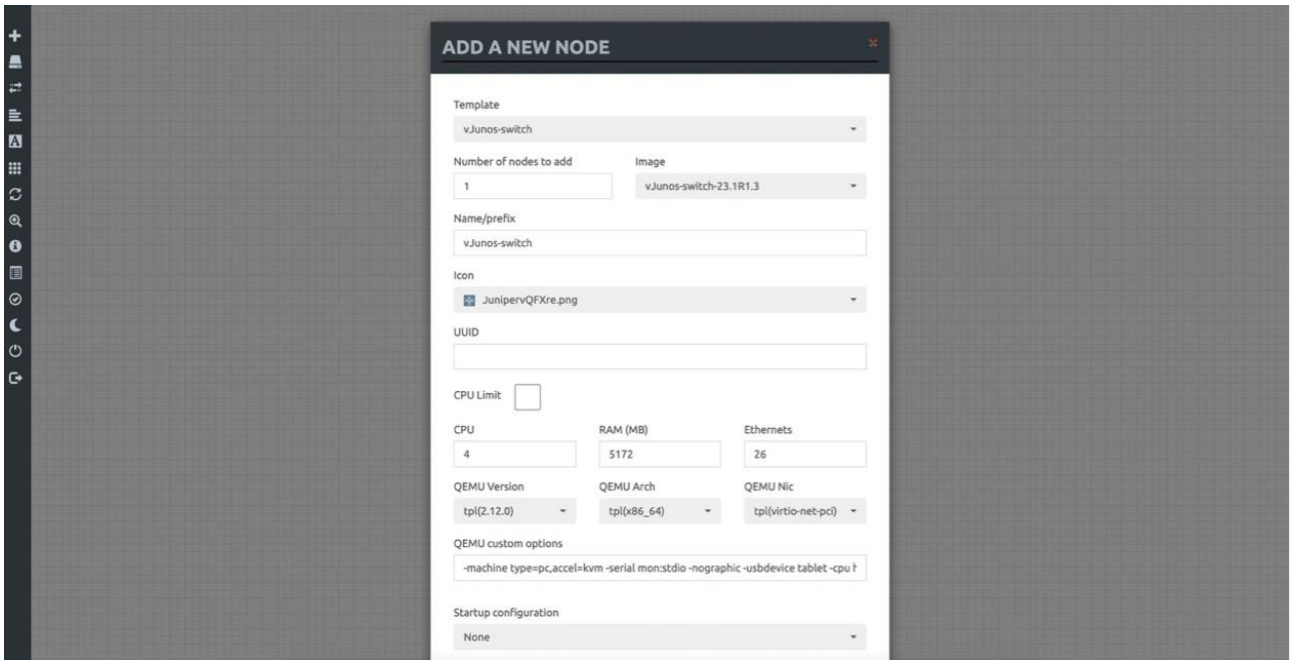
```
root@eve-ng:/opt/unetlab/addons/qemu/vJunosswitch-23.2R1# mv vjunos-switch-23.2R1.qcow2 virtioa.qcow2
root@eve-ng:/opt/unetlab/addons/qemu/vJunosswitch-23.2R1# /opt/unetlab/wrappers/unl_wrapper -a fixpermissions
```

On the EVE-NG UI, you should see ‘vJunosEvolved’ and ‘vJunos-switch’ as a device available to use and deploy:

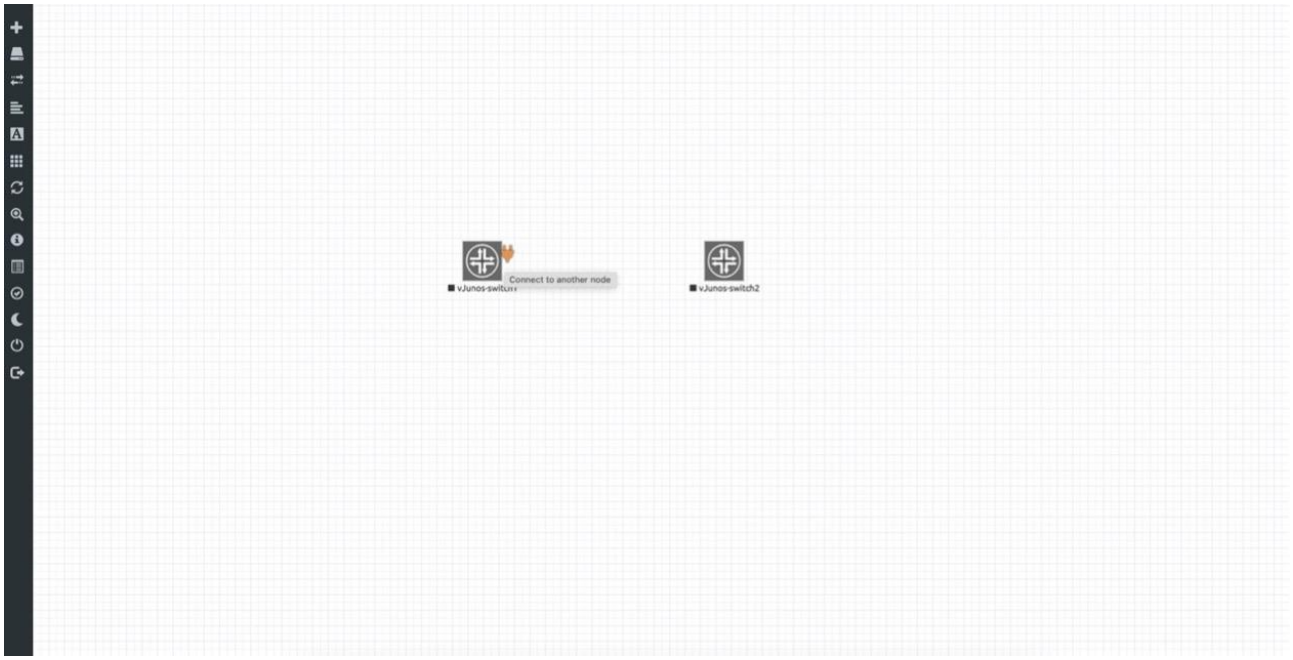
vJunosEvolved:



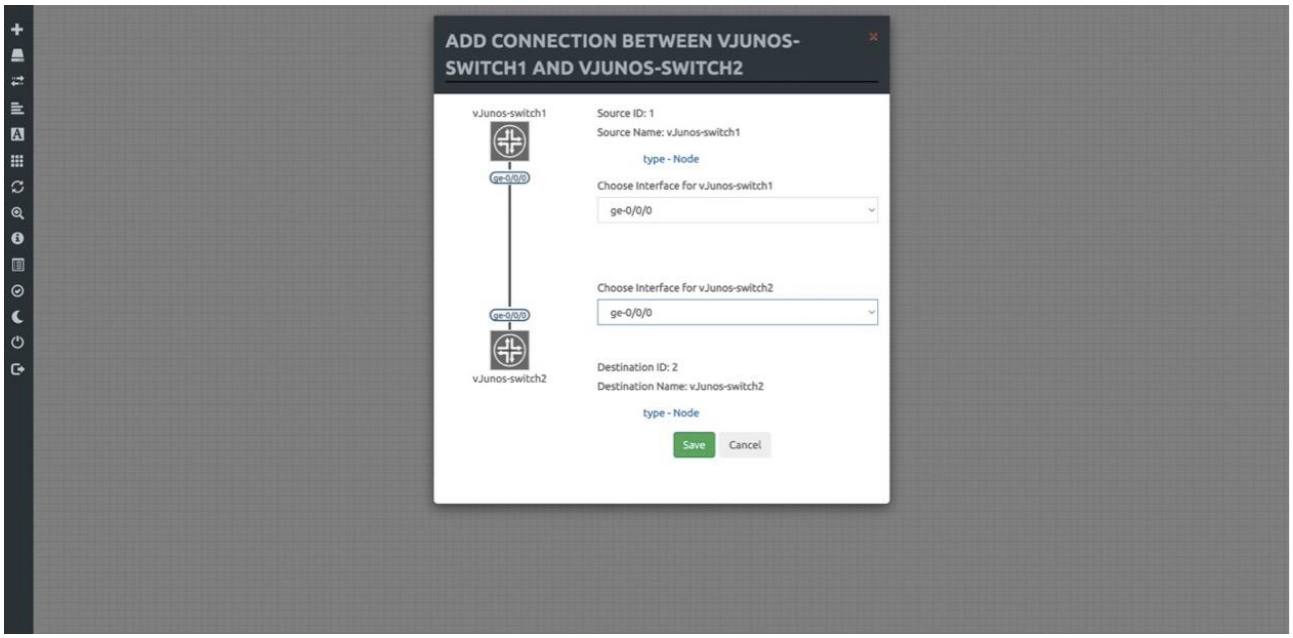
vJunos-switch:



These devices can now be deployed and interconnected. To interconnect two devices, simply click on the orange icon seen when you hover over a node, as seen below:

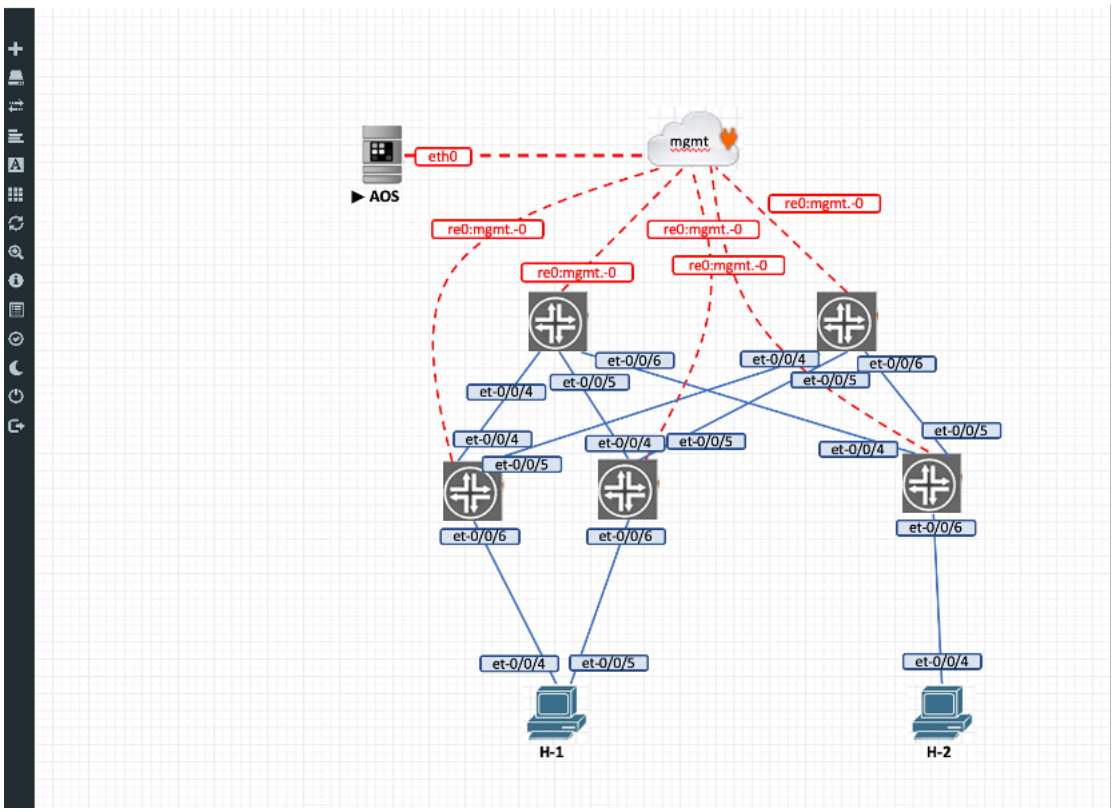


Once clicked, you can drag it to the destination node and let the connector go, which then gives you a pop-up to choose which interface is being connected on each side, like below:



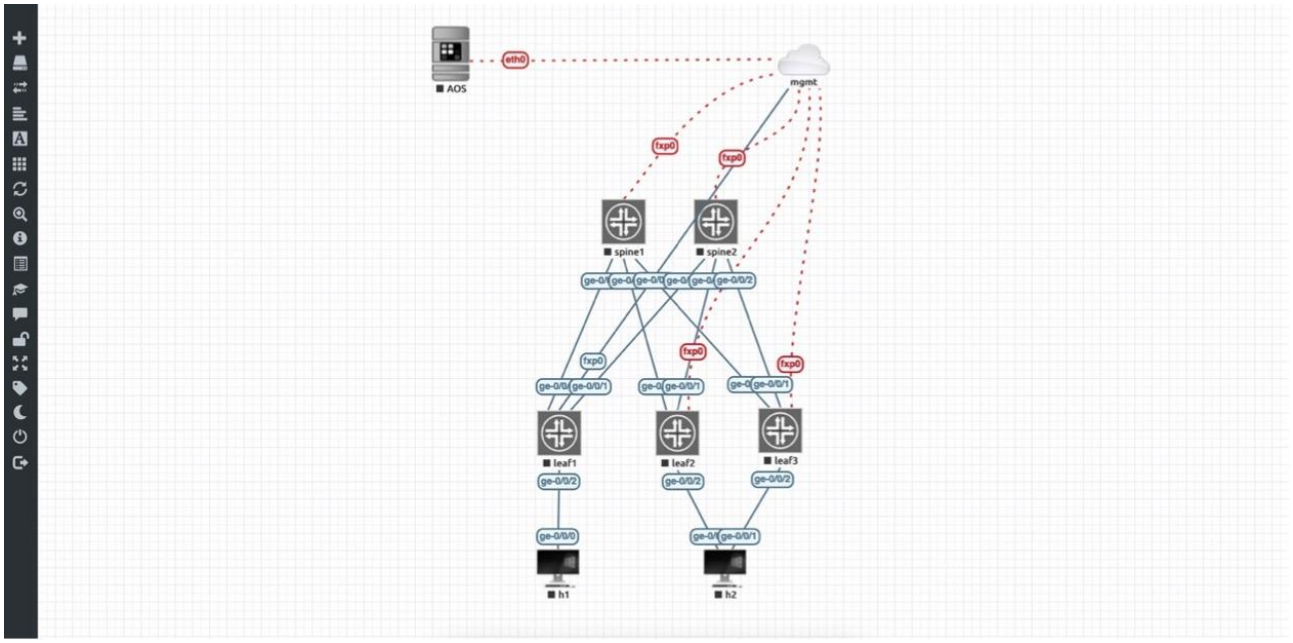
To demonstrate the use of vJunos-switch and vJunosEvolved, we've built the following topology on EVE-NG, respectively:

For vJunosEvolved:



The topology is a typical 3-stage Clos fabric, with two spines and three leafs. Two leafs (leaf1 and leaf2) are ESI peers and connect down to the same host, h1. The end goal is to manage these devices via Juniper Apstra and use Apstra to deploy all the necessary configuration to build a fabric and provide reachability between hosts h1 and h2, which are in the same subnet.

vJunos-switch:



This topology also is with two spines and 3 leafs. Two leafs (leaf2 and leaf3) are ESI peers and connect down to the same host, h2.

1.2.2 Deploying Juniper Apstra in EVE-NG

As seen in the topology above, we also have Juniper Apstra deployed within EVE-NG itself. For the sake of completeness, we'll also document how this is done.

The Juniper Apstra KVM image can be downloaded from the Juniper software downloads page - https://support.juniper.net/support/downloads/

The image is zipped with an extension of gz. Once unzipped, move the file to a folder created with the EVE-NG naming convention for Apstra – the directory must start with the prefix 'aos' and you can include the Apstra version as a suffix for easy identification of the image version. For example, we have created a folder named 'aos-4.1.2-269' and the image is under that. The image is finally renamed to 'hda.qcow2':

```
root@eve-ng:/opt/unetlab/addons/qemu# cd aos-4.1.2-269/
root@eve-ng:/opt/unetlab/addons/qemu/aos-4.1.2-269# ls -l
total 2762612
-rw-r--r-- 1 root root 2828908544 Feb 25 12:14 hda.qcow2
```

The following template can be used for Apstra:

"' root@eve-ng:/opt/unetlab/html/templates/intel# cat aos.yml



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